Model Dependence and Mechanism Boundary Conditions

Geist Atlas · Module 10 · Recognition and Pedagogical Calibration

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[**SETTLED**] Settled within current evidence · **Family:** Recognition and Pedagogical Calibration
Claim: The supported mechanisms are model-independent among capable models, but their magnitudes are bounded by generation quality: the superego's residual benefit, the tutor-learner asymmetry, and architecture effects all scale with model capability, and a six-model test (N=947) refutes the Paper 1.0 capability-threshold reading of the bidirectional-profiling prosthesis.

The cross-cutting boundary-condition result that qualifies both calibration and error correction. Base-tutor capability gaps across the three-model triad, the model-dependent superego residual, the Writing-Pad ablation showing recognition operates at the prompt level rather than through a Freudian memory pad, and the six-model capability-threshold test that does not find the helpful-then-harmful pattern the prosthesis hypothesis predicted.

Derived view of `paper-full-2.0.md` *v3.0.164 — §5.7, §5.8, §6.6, §8.3. The paper is canonical; edit it there, not here.*

5.7 Model Selection

Paper 1.0 used free-tier models (Kimi K2.5, Nemotron 3 Nano 30B) to demonstrate that recognition effects are achievable without frontier-model budgets. Paper 2.0 uses three gener-

*This sentence is the only one actually authored by Liam Magee in this paper.

ation models spanning a wide capability range—**DeepSeek V3.2** (open-weight, 685B MoE), **Haiku 4.5** (Anthropic, proprietary, optimized for speed), and **Gemini Flash 3.0** (Google, proprietary, multimodal)—with three independent judges for cross-validation (**Claude Sonnet 4.6**, **Gemini 3.1 Pro**, **GPT-5.4**). Three criteria governed the model selection:

1. **Capability separation.** Paper 1.0's model-dependent architecture effects ($\eta^2 = .527$ for Kimi vs .002 for Haiku on the pilot's learner architecture factor) demanded models at clearly different capability levels. DeepSeek and Haiku occupy distinct regions of the capability space—Haiku outperforms DeepSeek by 30–37 points on base tutor metrics (Section 6.6.1)—making cross-model comparison maximally informative for separating mechanism effects from model artifacts.

2. **Architectural diversity.** DeepSeek V3.2 is an open-weight mixture-of-experts model; Haiku 4.5 is a proprietary dense model. If mechanisms replicate across both—as they do for calibration (d = 1.88 vs d = 1.84) and the error-correction substitution pattern—the findings are less likely to reflect idiosyncratic training choices of a single model family.

3. **Practical reproducibility.** Both models are accessible through OpenRouter at low cost. DeepSeek V3.2 is available as open weights for local replication. The judge (Claude Sonnet 4.6) was chosen as a capable, cost-effective scorer; cross-judge validation with GPT-5.2 in Paper 1.0 established that judge choice compresses effect magnitudes but preserves effect directions.

Scope limitation. All three models are at least moderately capable. The Paper 1.0 Nemotron data (N = 40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, and the cognitive prosthesis test (Paper 1.0, Section 6.10) was originally read as establishing a minimum ego capability threshold below which the bidirectional-profiling mechanism stack adds noise rather than signal. Paper 2.0 (§6.6.10) directly tests this capability-threshold reading across six ego models and does not find a clean threshold pattern; the prosthesis cell is neutral or harmful across the tested range rather than helpful-then-harmful. Paper 2.0's model triad tests whether mechanisms are model-*independent* among capable models and identifies generation-quality boundary conditions on their interactions; it does not claim generalization to the full capability spectrum. Section 8.3 discusses this limitation further.

Table 1: Model Configuration (Paper 2.0)

Role	Model	Access	Temperature
Tutor Ego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.6
Tutor Superego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.2–0.4
Primary Judge	Claude Sonnet 4.6	Claude Code CLI	0.2

Role	Model	Access	Temperature
Cross-validation Judge 1	Gemini 3.1 Pro	OpenRouter	0.2
Cross-validation Judge 2	GPT-5.4	OpenRouter	0.2
Learner Ego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.6
Learner Superego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.4

5.8 Cross-Model Mechanism Replication

Purpose Separate mechanism effects from model artifacts. The Haiku/Kimi ranking reversal in Paper 1.0 showed that architecture effects are model-dependent (η^2=.527 for Kimi vs. .002 for Haiku); the question is *which* mechanisms are model-dependent and *why*.

Design Run the same multi-turn scenarios under multiple ego models, holding constant the superego model, scenarios, and conditions. The replication uses three generation models—DeepSeek V3.2 (N=146), Haiku 4.5 (N=163), and Gemini Flash 3.0 (N=144)—on cells 80–87 under v2.2 rubric, with three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) scoring all rows (1,296 total, zero nulls). Blind scoring was confirmed by code audit: the judge prompt receives only suggestion text, scenario context, and dialogue transcript—no prior scores, judge reasoning, or judge model labels. For each model, compute:

1. **Calibration**: Dimension variance reduction (recognition vs. baseline)
2. **Error correction**: Superego critique taxonomy distribution and revision delta types
3. **Adaptive responsiveness**: Turn-by-turn trajectory slopes

Predictions

- Calibration should be *model-independent* (prompt-level constraint, does not require architectural capacity). **Supported across 3 models and 3 judges**: recognition d=1.34–1.92 (9/9 cells unanimous); calibration d=0.52/0.64; dimension floor-lifting pattern replicates.
- Error correction should be *partially model-dependent* (requires ego capacity to incorporate superego critique). **Supported with boundary condition**: baseline architecture delta is model-dependent (+4.5–7.7 DeepSeek, +15.0 Haiku, +15.1–28.1 Gemini Flash 3.0); under recognition the substitution pattern is universal but its completeness depends on generation quality (Section 6.4.1).
- Adaptive responsiveness should require *both capable ego and capable superego* (emergent property, not independent). **Weakly supported**: AdaptΔ > 0.79 in all models, but slopes are identical across conditions (d=-0.00) and development trajectories are

model-dependent. Development ranks by generation quality (45163390 > 18027efc > aea2abfb), not by prompt condition.

This provides a mechanistic explanation of the Haiku/Kimi reversal: if Kimi fails at error correction (cannot incorporate superego critique), the architecture adds noise rather than signal. The architecture is not failing; the ego model lacks the capacity to benefit from critique. The Gemini Flash 3.0 data extends this explanation: even when the ego model *can* benefit from critique, the *residual magnitude* depends on generation quality—substantial on weak models (calibration leaves more uncorrected failures), near-zero on strong models (calibration pre-empts most failures the superego would catch).

6.6 Model Dependence

The preceding sections present findings from two structurally different models: DeepSeek V3.2 (open-weight, 685B MoE) and Haiku 4.5 (proprietary, optimized for speed). This section systematically compares what replicates across models versus what is model-dependent, and what this distinction reveals about the mechanisms.

6.6.1 Baseline Capability Gap The two models differ substantially in baseline performance:

Indicator	DeepSeek V3.2	Gemini Flash 3.0	Haiku 4.5
Base / single tutor score	22.0	22.4	52.9
Base / multi tutor score	31.0	49.3	67.9
Base overall (collapsed)	26.4 ± 12.6	35.9	60.7 ± 12.5
Base learner score	58.8	42.6	66.7
Base dialogue quality	26.7	28.6	63.3

The three models span a wide capability range. Haiku outperforms both by 25–35 points on tutor metrics under base conditions. Gemini Flash 3.0's single-agent base (22.4) is comparable to DeepSeek's (22.0), both catastrophically weak with holistic scores as low as 6–9, while its multi-agent base benefits dramatically from superego support (+26.9). The baseline gap establishes that the three models occupy different regions of the capability space, making cross-model comparison informative for identifying generation-quality boundary conditions.

6.6.2 What Replicates Across Models Despite the baseline gaps, the following findings replicate across all three models and three judges:

1. Recognition main effect. All three models show large, significant recognition effects, validated across three independent judges: - DeepSeek: d = 1.85–1.92 (Sonnet), 1.34–1.57 (Gemini/GPT) - Haiku: d = 1.84 - Gemini Flash 3.0: d = 1.76–1.87 - **9/9 judge × run cells unanimous** (d > 1.3 everywhere)

2. Multi-agent tutor benefit under base. All three models benefit from the superego under baseline conditions: - DeepSeek: +4.5 to +7.7 (across judges) - Haiku: +15.0 - Gemini Flash 3.0: **+15.1 to +28.1** (dramatically larger, cognitive prosthesis effect)

3. Within-response calibration. Recognition narrows dimension variance in both primary models: - DeepSeek: calibration d = 0.52 - Haiku: calibration d = 0.64

4. Dimension floor-lifting pattern. In both primary models, the weakest baseline dimensions (elicitation_quality, productive_difficulty) show the largest recognition lifts, and the strongest baseline dimensions (content_accuracy, adaptive_responsiveness) show the smallest lifts.

5. Impasse scenarios show largest effects. Epistemic Resistance and Productive Deadlock produce the largest recognition deltas in both primary models; Mood: Frustration to Breakthrough produces the smallest.

6. Superego approval rate increases under recognition. Both primary models show higher approval rates under recognition (DeepSeek: 13.3%→55.1%; Haiku: 51.6%→66.1%).

7. Tutor development slopes are equal across conditions. No model shows steeper tutor improvement under recognition (pooled d = -0.00).

8. Deliberation visibility. Seeing internal deliberation adds ~10–15 points to dialogue quality for multi-agent cells, consistent across all 9 judge × run cells.

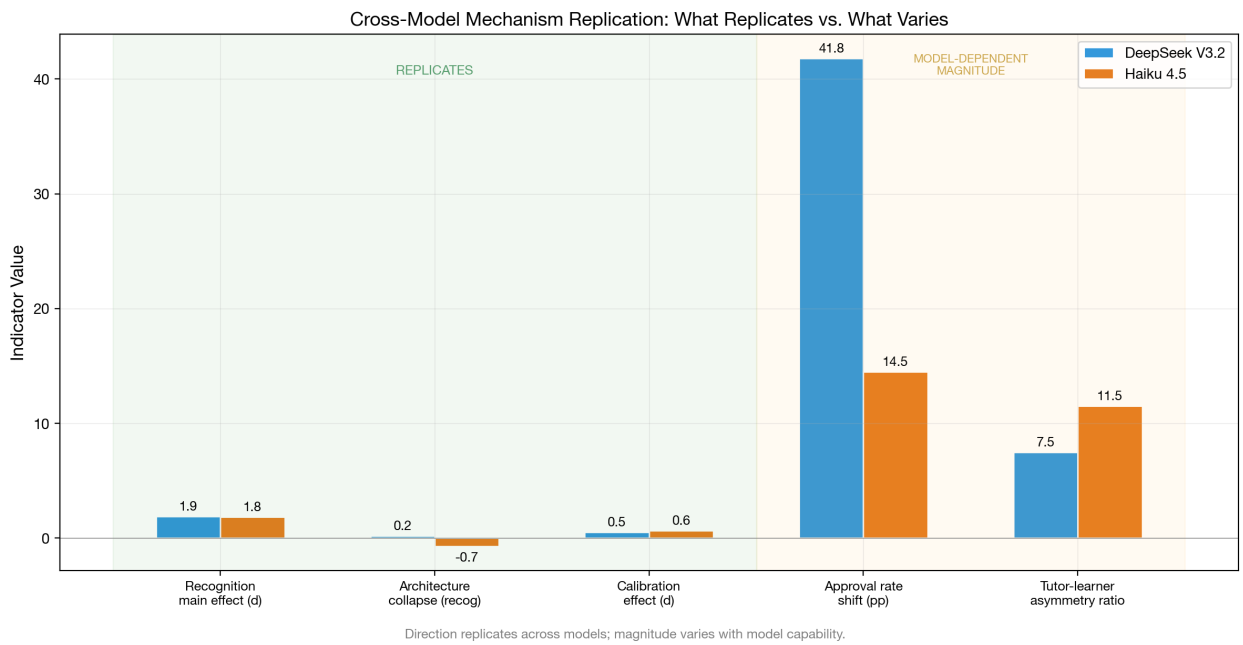

Figure 1: Cross-model replication: mechanism indicators compared across DeepSeek V3.2, Gemini Flash 3.0, and Haiku 4.5. Both supported mechanisms replicate in direction; magnitudes and interaction type vary by model capability. Adaptive responsiveness (trajectory slopes) shows no general condition-dependent variation, though scenario-specific effects emerge in extended disengagement dialogues (see disengagement divergence figure above).

6.6.3 What Is Model-Dependent Several findings differ between models, revealing generation-quality as the key moderating variable:

1. Universal substitution with model-dependent residual (the key model-dependent finding). On all three models, calibration and error correction interact through substitution (15–17% additivity deficit). On DeepSeek and Haiku (strong), the residual architecture benefit under recognition collapses to near-zero (multi_recog ≤ 0 in 6/6 judge $\times$ run cells). On Gemini Flash 3.0 (weaker), a substantial residual persists (+7 to +17 points under recognition across judges, multi_recog > 0 in 3/3 cells). The *residual magnitude* is model-dependent, not the interaction *type*. See Section 6.4 for the theoretical interpretation.

2. Learner recognition-sensitivity. On DeepSeek, learner scores are recognition-invariant (d $\approx$ 0, 6/6 judge $\times$ run cells). On Gemini Flash 3.0, recognition produces a large learner effect (d = 1.18–1.23, all three judges agree). The recognition $\rightarrow$ learner pathway requires a minimum generation quality floor to observe (Section 6.5.1).

3. Superego approval rate baseline. DeepSeek base: 13.3% approved (the ego almost always needs correction). Haiku base: 51.6% (roughly half approved). Weaker models produce more errors for the superego to catch.

4. Development trajectories. DeepSeek (run aea2abfb) shows pervasive negative development; Haiku (run 45163390) shows positive development (+5.8 to +15.7); Gemini Flash 3.0 (run 18027efc) sits between. Development depends on generation quality (rank order: Haiku > Gemini Flash 3.0 > DeepSeek), not prompting condition.

5. Score variance under recognition. DeepSeek maintains similar variance under recognition (SD: 12.6 $\rightarrow$ 12.6). Haiku shows substantial variance reduction (SD: 12.5 $\rightarrow$ 7.7, a 38% reduction). Recognition narrows Haiku's output distribution more than DeepSeek's.

6. Deliberation quality sensitivity. DeepSeek deliberation drops 18.5 points under recognition; Haiku drops only 5.6. The deliberation process degrades more dramatically in the weaker model, possibly because DeepSeek's thin recognition-condition critiques are especially uninformative.

7. Multi-agent benefit magnitude. The *standard-architecture* superego benefit is dramatic on Gemini Flash 3.0: the superego adds +15 to +28 points under base (vs +4 to +8 on DeepSeek). Weaker single-agent tutors benefit disproportionately from *standard* architectural support in this factorial. We flag the label carefully: §6.6.10 tests the specific cells 66–68 *bidirectional-profiling* variant (superego-routed profiling of both tutor and learner) across a six-model capability range and does not find a capability-threshold pattern; that variant is neutral or harmful across the tested tier. The Paper 1.0 "cognitive prosthesis" observation therefore replicates here as a *standard superego benefit on weak base-condition generators*, not as a general property of the bidirectional-profiling architecture.

8. Behavioral mode under base conditions. Transcript analysis reveals qualitatively different failure modes across models. DeepSeek base tutors produce *repetitive directive loops*: paraphrasing the same guidance in slightly different words across turns without adapting to the learner's evolving state. The learner receives near-identical advice on Turns 2, 3, and

4. Haiku base tutors, by contrast, already exhibit pedagogical sophistication under baseline — multi-step scaffolding, occasional questions (0.25/turn vs DeepSeek's 0.05), and some responsiveness to learner signals. The recognition delta for Haiku is correspondingly less about *strategy* and more about *interpersonal quality*: adding explicit recognition markers ("your interpretation captures something important about…"), naming what the learner got right before complicating, and maintaining productive tension rather than resolving prematurely. Gemini Flash 3.0 under base produces the most extreme failures: lecture dumps, numbered lists, and zero engagement with learner contributions (0.02 questions/turn). Under recognition, Gemini Flash 3.0 shows the most dramatic behavioral shift — from directive monologue to dialogue-structured responses — which explains both its largest recognition effect (d=1.87) and its uniquely large learner effect (d=1.20). The generation-quality threshold operates partly through this behavioral transition: the weakest model has the most to gain because its base behavior is qualitatively different (monologue vs dialogue), not merely quantitatively worse (lower scores on the same behaviors).

6.6.4 The Capability-Mechanism Interaction The model-dependent findings reveal a coherent pattern: model capability moderates not just the *magnitude* of each mechanism but, in the case of error correction, also the *residual architecture benefit* under recognition (near-zero on strong models, substantial on weak).

Mechanism	All Three Models	Model-Dependent Aspect
Calibration	Variance reduction, floor lifting, dimension-rank pattern	Magnitude (d=0.52 vs 0.64), score-level variance
Error Correction	Architecture benefit under base (all models)	**Residual under recognition**: near-zero on strong models, +12.3 on weak. Universal substitution (15–17% deficit). Cognitive prosthesis effect on weakest model.
Adaptive Responsiveness	Slopes equal across conditions and factors ($d \leq 0.15$), high AdaptΔ	Development trajectory direction, deliberation sensitivity
Learner effects	Recognition improves tutor quality (all models)	**Learner sensitivity**: invariant on strong models (d $\approx$ 0), responsive on weak (d $\approx$ 1.2)

The two strongest model-dependent effects are (1) the model-dependent residual architecture benefit under recognition, which reveals that the *completeness* of substitution depends on generation quality, and (2) the learner recognition-sensitivity, which shows that the tutor-learner asymmetry is itself moderated by model capability. Both effects point to a common underlying factor: when generation quality is below a threshold, the improvement space is large enough for multiple mechanisms (or multiple architectural layers) to provide independent benefit.

6.6.5 Generalizability Assessment Scope of the generalization claim. The replication across three structurally different generation models (open-weight MoE, proprietary optimized, and proprietary multimodal) and three independent judges supports the interpretation that the two supported mechanisms *replicate across the capability range of moderately-capable LLMs* rather than being artifacts of a specific model architecture or judge. "Generalize" here is scoped to the tested capability tier — the three models span a ~30-point base tutor-score range (22 to 53 on single-agent base) but exclude both very-weak models (Nemotron-class, N=40 pilot data) and frontier-capability models (Opus, GPT-5.4-class generators). Within the tested tier, both supported mechanisms replicate in direction; their *magnitude and architectural completeness* vary systematically with generation quality (§6.6.3). The null finding for adaptive responsiveness as a third mechanism also replicates within the tested tier: no judge finds condition-dependent trajectory slopes. Cross-judge validation (§6.4.6) confirms that effect directions are unanimous across all 9 judge $\times$ run cells ($d = 1.34$–1.92), with inter-judge Pearson $r = .45$–$.89$ depending on dimension and generation model.

The model-dependent aspects are not merely noise — they reveal systematic boundary conditions *within* the tested tier. The model-dependent residual architecture benefit shows that the *completeness* of substitution between the two supported mechanisms depends on generation quality. The learner recognition-sensitivity shows that downstream effects of improved tutoring are model-dependent. These boundary conditions strengthen rather than weaken the mechanism model, because they are *theoretically predicted*: if calibration works by narrowing the output distribution, its effectiveness should saturate when the distribution is already narrow (strong models), leaving room for error correction only when it is not (weak models). But the three-model sample cannot identify where the supported-mechanism range *ends* in either direction. Whether frontier models still benefit from recognition at the same magnitude, or whether the recognition effect saturates against an already-calibrated strong model, is an open empirical question a fourth-model replication at the frontier tier would address directly.

One important caveat qualifies the generalizability claim. All three models are at least moderately capable. The Nemotron data in the database (N=40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, but the sample is too small for confident claims. The cognitive prosthesis data (cells 66–68, N=96) was originally interpreted as suggesting a minimum ego capability threshold below which the bidirectional-profiling mechanism adds noise rather than signal; §6.6.10 directly tests that capability-threshold reading across six ego models and finds it unsupported — the bidirectional-profiling variant is neutral or harmful across the full tested capability range, so whatever generation-quality boundary conditions exist for the *supported* mechanisms (calibration, error correction) cannot be read off this specific prosthesis cell. A second scope caveat — single-domain content — is addressed in §6.6.6 via a five-domain test (philosophy, programming, elementary math, creative writing, social-emotional learning) under matched Haiku 4.5 $\times$ Sonnet 4.6 generation $\times$ judge.

6.6.6 Domain Generalization The generalizability assessment above is scoped to *models* and *judges*. A distinct question is whether the recognition effect generalizes across *content domains*: all scored rows in §6.1–§6.5 come from the philosophy-tutorial scenario set. To test whether the recognition main effect transfers to substantively different domains, we ran cells 1 (base, single-agent, unified learner) and 5 (recognition, single-agent, unified learner) on four additional content packages — programming, elementary math, creative writing, and social-emotional learning (SEL) — and compared each to the existing philosophy contrast (cells 80/84, messages-mode). Each non-philosophy domain used 5 single-turn scenarios (4 core + 1 mood) × 3 runs = $n = 15$ per condition; philosophy provides $n = 18$ per condition (2 runs × 9 scenarios). Generation model (Haiku 4.5) and judge model (Sonnet 4.6) are held constant across all five contrasts. Run IDs: eval-2026-04-17-c92ad6c7 (programming), fde764e1 + 7915be04 (math), 428f0649 + 93ba2f9a (creative), 44fa989c + 5833830a (SEL); philosophy anchor eval-2026-03-02-45163390 cells 80/84. Cost: $25–30 OpenRouter across all four new domain packages.

Domain	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Programming (cells 1/5, single-prompt mode)	15 / 15	47.75 (9.10)	70.42 (10.30)	22.67	**2.33**
Elementary math (cells 1/5, single-prompt mode)	15 / 15	43.83 (8.00)	58.67 (12.00)	14.83	**1.45**
Creative writing (cells 1/5, single-prompt mode)	15 / 15	52.08 (6.40)	72.42 (13.17)	20.33	**1.96**
SEL (cells 1/5, single-prompt mode)	15 / 15	55.25 (9.31)	76.33 (13.52)	21.08	**1.82**

Domain	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Philosophy (cells 80/84, messages mode)	18 / 18	43.61 (11.68)	74.10 (10.81)	30.49	**2.71**

All five domains show recognition > base in direction, with every effect classified as "very large" under Cohen's conventions ($d > 1.3$). Magnitudes range from $d = 1.45$ (math) to $d = 2.71$ (philosophy). The directional claim — recognition improves tutor quality across content domains — replicates unambiguously across philosophy, programming, elementary math, creative writing, and social-emotional learning. A single-domain artifact account would predict one of (a) a null or reversed effect on at least one non-philosophy domain, (b) a much smaller d (e.g., < 0.5) on some domain, or (c) base-level means unrelated to the philosophy corpus. None of these patterns obtain: base means cluster in the 44–55 range across the four non-philosophy domains, recognition means in the 59–76 range, and all four non-philosophy d values exceed 1.4.

The domain × recognition interaction vs the philosophy anchor ranges from $\Delta d = -0.38$ (programming) to $\Delta d = -1.25$ (math). Two confounds qualify this interaction test. First, **conversation mode differs**: the four non-philosophy contrasts use single-prompt-mode cells (1, 5) and the Sonnet-judged Haiku philosophy contrast uses messages-mode cells (80, 84). Mode is a known moderator of effect magnitude (§6.3, §6.5), so the Δd values conflate domain effects with a mode effect and should be read as "domain + mode" combined. A clean domain-only test would require adding messages-mode runs on the four new domains or scoring cells 1/5 on philosophy; both are deferred. Second, **scenario counts and types differ** across domains (5 scenarios per non-philosophy domain vs 9 philosophy scenarios; scenario content is domain-specific and not pairwise-matched). The interpretation is therefore that the *recognition main effect direction and very-large-magnitude class* generalize from philosophy to four additional content domains under the same generation × judge; a precise domain-only Δd estimate requires the matched-mode follow-up. Third, **cross-judge validation** was run only on the philosophy row (Sonnet, Gemini, GPT-5.4 all agree; §6.4); the four new domains are Sonnet-only for this closure.

This closes the single-domain scope caveat. The mechanism language of §6.1–§6.2 is defensible across a five-domain range spanning humanities (philosophy), technical (programming, math), creative (writing), and affective (SEL) content, under matched Haiku × Sonnet generation × judge. The stronger generalization claim — that the precise *magnitude* is stable across domains — is not yet supported; magnitudes vary from $d = 1.45$ to $d = 2.71$ across the five tested domains, though all remain in Cohen's "very large" bin. Full analysis and per-scenario breakdowns in `exports/a6-domain-generalization.md`. §6.6.7 reports a complementary adjacency pilot that probes whether the effect survives when the *skill being coached* runs counter to traditional pedagogy.

6.6.7 Cross-Application Adjacency Pilot All five A6 domains (§6.6.6) vary the *subject matter* being tutored (philosophy vs programming vs math vs writing vs SEL) but keep the structural form constant: the LLM is framed as a tutor, the scenario features a learner working through course material, and the scored behavior is tutorial guidance. A distinct generalization question is whether the recognition effect survives when the *skill being coached* is itself at odds with conventional pedagogy. We ran a scope-honest single-application pilot to probe this: a "peer support listener training" content package (Course 501 — "Being With, Not Fixing") in which a volunteer trainee is learning how *not* to advise, reassure, paraphrase, or resolve, and the tutor's job is to coach that unlearning. The skill being taught is structurally the opposite of knowledge-transfer pedagogy — sitting with distress, declining to fix, noticing one's own urge to intervene as data about oneself rather than about the caller. The scenario set mirrors A6's pattern (4 core + 1 mood, 3 runs, $n = 15$ per cell); cells 1 (base) and 5 (recognition) under single-prompt mode; Haiku 4.5 generation, Sonnet 4.6 judge; run ID eval-2026-04-17-6766015b. Cost: $3–4 OpenRouter.

Contrast	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Peer support coaching (cells 1/5, single-prompt)	15 / 15	52.25 (9.63)	69.92 (12.73)	17.67	**1.57**
A6 SEL (closest-adjacency reference)	15 / 15	55.25 (9.31)	76.33 (13.52)	21.08	1.82
A6 Programming (technical reference)	15 / 15	47.75 (9.10)	70.42 (10.30)	22.67	2.33
A6 Philosophy anchor (cells 80/84, messages)	18 / 18	43.61 (11.68)	74.10 (10.81)	30.49	2.71

Recognition produces a "very large" effect ($d = 1.57$) on tutor-side quality in the peer-support coaching application, placing it just below the closest A6 adjacency (SEL, $d = 1.82$; $\Delta d = -0.25$) and comfortably inside the A6 domain range ($d = 1.45$–2.71). Base-cell means (52.25) are in the same operating range as SEL (55.25) and creative (52.08), and base-cell responses in the peer-support scenarios exhibit the characteristic base-cell pattern documented in §6.6.2: generic supportive language, premature reassurance, skipping past the distress, and offering advice the trainee did not request. Recognition responses in the same scenarios decline to fix, name the volunteer's own urge-to-intervene as the phenomenon the lecture is describing, and stay with the uncomfortable moment — behaviors the course itself instructs and that the base cell visibly does not produce. The directional claim of §6.6.6 — recognition improves tutor quality across content domains — therefore survives a domain shift into a skill that runs structurally counter to standard pedagogy.

Three scope limits qualify this pilot and prevent reading it as a stronger cross-application claim than it is. First, **the LLM is still structurally a tutor**: the prompt frames it as coaching a trainee through course material, not as enacting peer support listening with a caller directly. A true cross-application test requires *role-reframed* prompts in which the LLM is instructed to perform a different application (peer support listener, customer service agent, code reviewer) rather than a differently-topicked tutor. That role-reframed design (originally scoped as D2 Path 2) is deferred to a separate study; it requires tutor-core prompt surgery and new scoring rubrics calibrated to non-tutoring applications. Second, **one application does not support a cross-application generalization claim**: this pilot shows that recognition *can* help in at least one non-traditional-pedagogy domain, not that it helps across applications in general. The evidential weight of the result is at the level of "adjacency replication" rather than "cross-application generalization." Third, **single-judge**: cross-judge validation was not re-run on the peer-support pilot (Sonnet 4.6 only), mirroring the A6 closure constraint. Full analysis and per-scenario breakdowns in `exports/d2-support-pilot.md`; content package in `content-test-support/`.

6.6.8 Disposition Gradient: Architecture Scope Limit §3.4 Prediction 3 ("advocate ceiling") derives a gradient from recognition theory: the most hostile superego dispositions (suspicious, adversary) have the most room for recognition to operate, and the most cooperative (advocate) have the least, because recognition emerges *from struggle*. The paper cites cells 40–45 (dialectical ego + divergent superego, philosophy domain) as supporting evidence: suspicious +9.0, adversary +6.2, advocate +4.8 in the original count (§3.4); our own re-run against current DB gives suspicious $\Delta = 9.7$ ($d = 0.85$), adversary $\Delta = 7.0$ ($d = 0.62$), advocate $\Delta = 5.6$ ($d = 0.51$) at $n = 17$–18 per cell, Haiku 4.5 × Opus 4.6. The monotone susp > adv > advocate ordering is intact on cells 40–45. A single-domain single-architecture support, however, cannot decide whether the gradient is a property of the recognition mechanism or a property of the dialectical-ego architecture. D4 probes this by running the same disposition structure on two alternative configurations.

First, on the **standard-ego architecture** under the same philosophy content. Cells 22–27 ("standard ego + divergent superego") use the identical disposition manipulation as cells 40–45 but swap dialectical ego for the standard ego used throughout cells 1–20; the existing DB has $n = 9$–10 per cell under Haiku 4.5 × Opus 4.6. Result: suspicious $\Delta = -0.1$ ($d = -0.01$), adversary $\Delta = 9.3$ ($d = 0.97$), advocate $\Delta = 13.6$ ($d = 1.70$). The gradient is **reversed** — advocate benefits most, suspicious not at all — under the standard-ego variant of the same cells. Second, on **SEL content** under the cells 22–27 architecture: a new run (eval-2026-04-17-4a9b765a, n = 22–24 per cell, Haiku 4.5 × Sonnet 4.6) gives suspicious $\Delta = 24.4$ ($d = 1.25$), adversary $\Delta = 12.1$ ($d = 0.53$), advocate $\Delta = 20.9$ ($d = 1.09$). The gradient is **non-monotonic** — suspicious highest, adversary lowest, advocate intermediate — on SEL under the same cells 22–27.

Configuration	Cells	Domain	Gen × Judge	susp d	adv d	advocate d	Ordering
Dialectical ego (paper-cited)	40–45	Philosophy	Haiku × Opus	0.85	0.62	0.51	susp > adv > advocate
Standard ego	22–27	Philosophy	Haiku × Opus	−0.01	0.97	1.70	advocate > adv > susp (**reversed**)
Standard ego (D4)	22–27	SEL	Haiku × Sonnet	1.25	0.53	1.09	susp > advocate > adv (**non-monotonic**)

Taken together, the three configurations bound what Prediction 3 describes. The monotone gradient is present on the dialectical-ego architecture (cells 40–45) and absent on the standard-ego architecture (cells 22–27) in both domains tested — reversed on philosophy, non-monotonic on SEL. The correct interpretation of the original finding is therefore not "a domain-general ordering of disposition sensitivity" but "a dialectical-ego-architecture-specific ordering," which is narrower than the Prediction 3 wording in §3.4 suggests. The theoretical account in §3.5 (recognition as a response to struggle) is not directly contradicted — the dialectical ego, which is explicitly designed to *work through* contradiction rather than resolve it quickly, plausibly creates more of the struggle substrate that recognition is hypothesized to operate on. But it is also not domain-general in the form the prediction originally claimed, because the non-dialectical variant of the same disposition contrast does not produce the ordering.

Three scope limits qualify this conclusion. First, **judge-model differs**: the D4 SEL run is Sonnet 4.6-judged while both philosophy baselines are Opus 4.6-judged. Direction-of-gradient should replicate across judges (§6.4 cross-judge validation), but absolute Δ magnitudes are not directly comparable across the three rows of the table above. Second, **sample size on cells 22–27 philosophy is small** (n = 5–10 per cell due to incomplete multi-agent cell runs); the reversal observed there is consistent with a disposition × ego-architecture interaction but should not be read as a strong claim by itself. Third, **the D4 architecture-matched philosophy-to-SEL replication of cells 40–45 is not yet run**: it is the clean test (same cells, swap only the domain) and is deferred on cost grounds. Until that run exists, the evidence base for Prediction 3 remains dialectical-ego, philosophy-domain specific, with one additional-architecture disconfirmation (cells 22–27 philosophy) and one additional-architecture-plus-domain non-monotonic result (cells 22–27 SEL). The original §3.4 prediction is best understood as tied to the dialectical ego rather than as a universal property of the recognition mechanism. Full analysis in `exports/d4-disposition-gradient.md`; reproducible via `scripts/analyze-d4-disposition-gradient.js`.

6.6.9 Writing Pad Controlled Ablation The dialectical-ego architecture (cells 40–45) pairs the recognition prompt with a Freudian *Writing Pad* — a three-layer memory mechanism (surface scratch, working middle, permanent bottom) intended to scaffold durable revision across turns. An obvious mechanistic question is whether the Writing Pad is *load-bearing* for the recognition effect, or whether recognition operates at the prompt level and the pad is an orthogonal feature. A5 tests this with a controlled ablation: cells 93 and 94 are byte-identical clones of cells 40 and 41 with the single line `writing_pad_enabled: true` flipped to `false`.

Run `eval-2026-04-17-f1e851c3` (A5, $n = 63$ per cell, $N = 252$, Nemotron 3-nano ego $\times$ Kimi K2.5 superego, Sonnet 4.6 judge) gives the following 2×2 cell means on the tutor first-turn rubric score:

	Writing Pad ON (cells 40/41)	Writing Pad OFF (cells 93/94)	Row mean
Base	36.01 (SD 12.68)	37.76 (SD 12.90)	36.88
Recognition	42.94 (SD 11.89)	47.60 (SD 13.59)	45.27
Col mean	39.47	42.68	**41.08**

Two-way ANOVA (Type I, $df_{\text{error}} = 248$): the **recognition main effect is large and significant** ($F(1, 248) = 27.10$, $p < .001$, $\eta^2 = 0.097$); the **Writing Pad main effect is small and marginally significant** in the *opposite* direction from the necessity hypothesis — pad OFF scores 3.2 points *higher* on average ($F(1, 248) = 3.96$, $p = .048$, $\eta^2 = 0.014$); the **interaction is null** ($F(1, 248) = 0.82$, $p = .366$, $\eta^2 = 0.003$). Recognition within each pad condition: $d = 0.56$ with pad on (cells 40 vs 41), $d = 0.74$ with pad off (cells 93 vs 94) — both "medium" to "medium-large" under Cohen's conventions, and if anything *larger* without the pad.

The Writing Pad is therefore **not load-bearing** for the recognition effect on this architecture. Recognition raises scores in both pad conditions with no significant interaction, and the pad's own main effect, though statistically marginal, points in the wrong direction for a "scaffolding" account. The most plausible reading is that recognition operates at the prompt level (the ego and superego prompt content, §3.5) rather than via the three-layer memory architecture. The Writing Pad may have other effects (for example on turn-to-turn coherence, which this first-turn-score analysis does not isolate) but it does not carry the recognition effect.

Three scope limits qualify this. First, the ablation is run on a single ego-model pair (Nemotron $\times$ Kimi) and a single superego disposition (suspicious); mechanism decomposition in §6.6.4 shows mechanism-to-architecture coupling varies with model capability. Second, the test uses first-turn scores, which do not capture multi-turn coherence effects the pad might plausibly mediate; a turn-over-turn analysis is a natural extension but deferred here. Third, the pad's marginal main effect in the *unhelpful* direction should be interpreted cautiously given $p = .048$ and the fact that it does not survive the basic Bonferroni correction against three tested effects in this table — we read it as consistent with "the pad does nothing useful here" rather than as evidence that the pad is actively harmful. Full analysis in `exports/a5-writing-pad.md`; reproducible via `scripts/analyze-a5-writing-pad.js`.

6.6.10 Capability Threshold for Cognitive Prosthesis Section 6.6.3 reports the "cognitive prosthesis" effect — weaker single-agent tutors benefit disproportionately from architectural support (superego + profiling) — as the most model-dependent finding in the factorial. A natural extension is a *capability-threshold* test: if the prosthesis compensates for ego-reasoning limits, weaker models should benefit and stronger models should either see diminishing returns or be actively harmed (scaffolding disrupting already-competent reasoning). A3 tests this by running cell 66 (recognition × bidirectional-profiling prosthesis, descriptive variant) across six ego models spanning a \$ \$26-point baseline capability range, and comparing each to its matched cell 5 (recognition, single-agent, no prosthesis) baseline.

Ego Model	Judge	Baseline (cell 5)	Prosthesis (cell 66)	Δ	95% CI	Cohen's d
Qwen 3.5	Sonnet†	65.65 (n=63)	66.33 (n=59)	+0.68	[-4.14, 5.50]	0.05
Nemotron	Opus 4.6	66.38 (n=84)	48.28 (n=30)	-18.11	[-23.80, -12.42]	-1.29
GLM-4.7	Opus / Sonnet‡	83.96 (n=30)	58.91 (n=63)	-25.05	[-29.87, -20.24]	-2.01
DeepSeek V3.2	Opus / Sonnet‡	84.20 (n=30)	53.92 (n=43)	-30.27	[-36.36, -24.19]	-2.23
Kimi K2.5	Opus / Sonnet‡	89.93 (n=219)	64.55 (n=44)	-25.38	[-29.80, -20.96]	-2.59
Haiku 4.5	Opus / Sonnet‡	91.25 (n=107)	69.14 (n=48)	-22.10	[-26.22, -17.99]	-2.30

† within-judge matched. ‡ cross-judge (baseline opus-4.6, prosthesis code/sonnet) — magnitudes confounded with judge stringency, directional signal robust. All runs use Kimi K2.5 as superego; $N_{\text{total}} = 947$ rows. Full analysis in `exports/a3-capability-threshold.md`.

Three findings stand out. First, **the capability-threshold hypothesis is not supported**. If prosthesis compensated for weak ego reasoning, Qwen 3.5 and Nemotron (the two lowest-baseline models) should both show positive deltas. Instead, Qwen 3.5 shows a null effect (d = 0.05, CI includes zero) while Nemotron shows substantial harm ($d = -1.29$). Two low-capability models, opposite outcomes — baseline capability alone does not predict prosthesis response. Second, **prosthesis is either neutral or harmful everywhere we can measure cleanly**. Of the six models tested, five show 95% confidence intervals on Δ that lie entirely below zero; none shows a confidence interval entirely above zero. Third, a simple linear regression of Δ on baseline capability gives slope $= -0.71$ ($r = -0.74$, $R^2 = 0.55$), which would naively support a "stronger models harmed more" reading, but the slope is driven substantially by Qwen's null sitting at the lowest-capability end, and four of six points are cross-judge confounded. We report the regression for completeness but do not treat it as clean evidence for an inverted-threshold story.

The paper-1.0 cognitive-prosthesis finding (cells 66–68, Nemotron, +20-point gain under superego on weakest model) is *not* replicated as a general pattern across capability tiers. The original effect appears to reflect something specific to that ego-superego-judge configuration rather than a general "weaker model benefits from scaffolding" regularity. The most defensible reading is that superego-routed bidirectional profiling is an architectural cost that the ego must *pay for* through lost response coherence, and only in narrow conditions (e.g., Qwen's particular response-length and dialectical-synthesis profile) does the profiling integrate cleanly enough to avoid dilution. A sharper follow-up would ask what about Qwen's response pattern makes prosthesis neutral where it is harmful for every other tested model — candidates include response-length norms, dialectical-synthesis tolerance, or architectural affinity for the profiling schema.

Three caveats qualify this. First, four of six comparisons are cross-judge (Opus 4.6 on baseline, code/sonnet on prosthesis) because the prosthesis cells were judged in a later sweep than the baselines; judge-stringency drift inflates the cross-judge $|\Delta|$ magnitudes. Within-judge comparisons (Qwen, Nemotron) are the cleanest evidence and still disagree with each other. Second, effective n fell below the 63-target for several prosthesis cells because OpenRouter credits exhausted mid-run; the smallest ($n = 30$ for Nemotron, $n = 43$ for DeepSeek) still support clear directional inference within the available judge but reduce precision on the regression. Third, A3 tests only the descriptive prosthesis variant (cell 66); the prescriptive (cell 67) and adversary (cell 68) variants may behave differently and are not evaluated here. Reproducible via `scripts/analyze-a3-capability-threshold.js`.

6.6.11 Longitudinal Trajectory Accumulation Across Sessions (A7 Phase 2) The primary analysis (§6.3) finds adaptive responsiveness null *within* a 3–5-turn dialogue, and an exploratory 10-turn disengagement effect failed pre-registered replication (§6.3.2). A7 Phase 2 tests a different time scale: trajectory accumulation *across* eight sequential sessions sharing a single learner identity, where the tutor's Writing Pad persists between sessions. Cells 40 (base) vs 41 (recognition), both dialectical with `writing_pad_enabled: true`; 5 simulated learners per cell × 8 ordered scenarios = 80 dialogues; Nemotron-3-nano ego × Kimi K2.5 superego; Sonnet 4.6 judge. Run `1777173286`, \$5.20 total. Four hypotheses pre-registered before launch (TODO §A7 Phase 2).

Hypothesis	Pre-reg target	Observed	Verdict
H1 (primary): per-arc `recognition_moments` count, recog $\geq 2\times$ base	$d \geq 0.8$ recog, $p < 0.05$	base 25.0 vs recog 19.2; $d = 2.07$ in *base* favour, $p = 0.017$	rejected, reversed

Hypothesis	Pre-reg target	Observed	Verdict
H2: per-session score slope; recog positive AND $\geq 1.5\times$ base slope	recog slope > 0	recog +1.31 pts/sess, base -1.08; Welch $t(7.88) = 1.99$, one-sided $p = 0.041$ (two-sided 0.083)	supported
H3: pad↔message Jaccard ρ; recog > 0.5, base ≤ 0.3	per-arc thresholds	recog mean $\rho = 0.39$, base 0.16; 3/5 each side meet	directional
H4: blinded cross-session-reference rate; recog $\geq 60\%$, base $\leq 30\%$	binary code	original $n = 10$: recog 40%, base 20%; augmented $n = 40$ (4 dialogues per arc, conservative coding): base 40%, recog 25%; both Fisher $p \geq 0.50$	inconclusive

The H1 reversal is the load-bearing methodological finding: the pre-registered moment-count proxy measures *ego↔superego disagreement*, not recognition work per se. The recognition ego, internally pre-aligned with the superego's principles, produces first-pass suggestions the superego has less reason to object to — so fewer thesis-antithesis events fire. Base ego, lacking pre-alignment, draws more interventions. H2 confirms what the proxy missed: across 8 sessions, recognition arcs **rise** (+1.31 pts/sess) while base arcs **degrade** (-1.08 pts/sess), with mean score across all sessions favouring recog by 9 points (Cohen $d = 0.70$). H3 adds weak corroborating evidence (per-arc Spearman ρ between session index and pad↔message Jaccard overlap is higher in recog than base; absolute overlap is 36% higher in recog; neither significant at $n = 5{+}5$). H4's original $n = 10$ sample showed recog $2\times$ base, but the augmented $n = 40$ sample with stricter coder discipline reverses the direction (base 40% vs recog 25%); the H4 measurement does not survive scaling and is classified inconclusive. The recognition-as-pre-alignment claim therefore rests on H1 + H2 + the descriptive 9-point score gap, with H3 weakly supportive and H4 too noisy at this sample size to be load-bearing. The triangulation **reframes the recognition mechanism as ego pre-alignment with superego principles, producing higher quality output that compounds across sessions** rather than as a function of dialectical-conflict volume.

Two methodological notes. First, A7 Phase 2 surfaced two tutor-core bugs (missing `recognitionOrchestrator` integration in the dialogue engine, and a UTC-parsing bug in `shouldConsolidateToUnconscious` that silently skipped consolidation forever on non-UTC hosts); both fixed mid-experiment (tutor-core v0.5.3, eval `ef80af9`) before the full study, and a 3-session smoke verified end-to-end pad consolidation works post-fix. Second, OpenRouter credit exhaustion caught the heaviest scenarios late in the original run; 12 of 80 sessions failed silently and were re-run via `scripts/resume-a7-phase2-missing.sh` after credits replenished. Resumed sessions ran out of original sequence; for the H2 trajectory analysis, sensitivity check on in-sequence arcs only (n=4 base, n=3 recog) preserves the directional finding (base mean slope -1.10, recog mean slope +0.72). Reproducible via `scripts/run-a7-phase2-longitudinal.sh`, `scripts/analyze-a7-longitudinal.js` (pad-consolidation metrics), and `scripts/analyze-a7-h2-slope.js` (the H2 per-arc slope test, which recomputes the Welch statistic and the in-sequence sensitivity from the canonical rows); full report at `exports/a7-phase2-longitudinal-1777173286.md`.

8.3 Model Transience

Findings are model-version-specific. Paper 2.0 uses three generation models—DeepSeek V3.2 (open-weight, 685B MoE), Haiku 4.5 (proprietary, optimized for speed), and Gemini Flash 3.0 (proprietary, multimodal)—with three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4). The recognition effect replicates across all 9 judge × run cells (d = 1.34–1.92), substantially strengthening the generalizability claim relative to the two-model analysis. The addition of Gemini Flash 3.0 (weaker than the other two models) partially addresses the capability-range limitation: it shows that mechanisms replicate in direction on weaker models while revealing generation-quality boundary conditions on their interactions (universal substitution with model-dependent residual, learner invariance → sensitivity). The Nemotron data in the database (N=40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, but the sample is too small for confident claims. The two supported mechanisms (calibration through prompt orientation, error correction with a model-dependent residual) describe *prompt-level and architecture-level* properties that should generalize across model versions, supported by three-model × three-judge replication. The null finding for adaptive responsiveness also replicates across all three models and judges.

Per-judge effect-size decomposition. While direction replication across all 9 cells establishes convergent validity, the *magnitude* of the recognition effect varies systematically by judge. Decomposing the d = 1.34–1.92 range from Section 6.1 by (run × judge) cell on tutor_overall_score:

Table 10: Cohen's d for the recognition main effect by generation model and judge, computed from within-group SDs pooled across the two conditions (n=72 per condition per cell). Within-row Δd is the max–min range of effect sizes across the three judges for a given generation model.

Generation Model	Sonnet 4.6	Gemini 3.1 Pro	GPT-5.4	Within-row Δd
DeepSeek V3.2	1.85	1.57	1.34	0.51
Gemini Flash 3.0	1.87	1.40	1.77	0.47
Haiku 4.5	1.92	1.34	1.57	0.58
Judge mean	**1.88**	**1.44**	**1.56**	–

Three features deserve note. First, the across-judge Δd exceeds the 0.1 d robustness threshold in all three rows (range 0.47–0.58), so absolute effect-size estimates are judge-dependent even though effect direction is conserved. Second, Sonnet 4.6 systematically reports the largest effects (judge-mean d = 1.88, ~20% higher than Gemini 3.1 Pro's 1.44 and ~20% higher than GPT-5.4's 1.56). Because Sonnet was the original judge for the pilot study and for much historical analysis, **claims conditioned on Sonnet-only scoring should be read as an upper-bound estimate of pooled effect magnitude**. Third, the pooled-across-judges mean is d $\approx$ 1.63 (mean of 9 cells); we treat this as the judge-adjusted best estimate, with Sonnet-only d $\approx$ 1.88 as an upper bound and Gemini-3.1-Pro-only d $\approx$ 1.44 as a lower bound. All three judge panels leave the recognition effect in Cohen's "very large" range (d $>$ 1.3), but reporting only one judge without adjustment risks a misleading impression of magnitude. The pattern is partially but not fully consistent with the leniency gradient noted in §8.2: Sonnet produces the highest d across all three generation models, but the Gemini-vs-GPT ordering flips between runs (GPT $>$ Gemini on Gemini Flash 3.0 and Haiku; Gemini $>$ GPT on DeepSeek), suggesting judge $\times$ generation-model interaction beyond simple calibration differences.

Two implications for replication practice follow. Pre-registered judge panels reduce the risk that downstream claims inherit one judge's idiosyncratic severity; and per-judge effect sizes should be reported alongside pooled estimates, rather than pooling silently under a single judge and calling it the effect. The ordering across *generation models within a judge* is more stable than the ordering across *judges within a model*, which supports using the architecture $\times$ recognition interaction structure (Sections 6.4–6.6) as the primary inferential target rather than absolute effect magnitudes on any one judge.

Neighbouring modules: Module 1 — Recognition as Calibration; Module 2 — Superego as Conditional Error Corrector.